

FUNCTION SPACES

Soumyashant Nayak, notes by Ramdas Singh

Fifth Semester

Contents

1	FUNCTION SPACES	1
1.1	An Introduction	1
1.2	Metric and Normed Spaces	2
	Index	5

Chapter 1

FUNCTION SPACES

July 20th.

Quizzes 1 and 2 will account for 10% and 15% of the final grade respectively. The mid-term will account for 25% of the final grade, and the final exam will account for 50%.

1.1 An Introduction

Suppose $\mathbb{K}$ is a field, and X is a topological space. Then the set of all $\mathbb{K}$ -valued functions on X , denoted by $\mathbb{K}^X$, is a vector space over $\mathbb{K}$. If we let $X = \{1, 2, \dots, n\}$, then $\mathbb{K}^X \cong \mathbb{K}^n$, the n -dimensional vector space over $\mathbb{K}$. In this course, we will be primarily concerned with the case when $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Consider the map

$$D : \mathbb{K}^{\{1, 2, \dots, n\}} \rightarrow \mathbb{K}^{\{1, 2, \dots, n-1\}}, \quad (Df)(k) = f(k+1) - f(k) \text{ for } 1 \leq k \leq n-1, \quad (Df)(n) = f(1) - f(n). \quad (1.1)$$

Here, D acts as a discrete analogue of the derivative operator by taking finite differences. We can split this finite difference map as the difference of the left shift operator and the identity operator as $D = S - I$. Since S is the left shift operator, we have $S^n = I$ so the eigenvalues of S are a subset of the n -th roots of unity. Let λ be such an eigenvalue. Then a corresponding eigenvector can be given as

$\begin{pmatrix} 1 \\ \lambda \\ \vdots \\ \lambda^{n-1} \end{pmatrix}$. If $\lambda = e^{2\pi i/n}$, then we can denote the corresponding eigenvector of λ^k as χ_k . In fact, in the

basis $\{\chi_1, \dots, \chi_{n-1}, \chi_n\}$, D is a diagonal matrix. Such a basis is termed a *discrete Fourier basis*, and the coordinate vector of D with respect to this basis is called the *discrete Fourier transform* of D . The Fourier transform diagonalizes the finite differential operator. We can view χ_k as a function $\chi_k : \{1, 2, \dots, n\} \rightarrow \mathbb{C}$ defined by $\chi_k(j) = e^{2\pi i k(j-1)/n}$ for $1 \leq j \leq n$.

Recall the Hermitian inner product on $\mathbb{C}^n$ given by $\langle x, y \rangle = \sum_{i=1}^n x_i \overline{y_i}$. Viewing $\mathbb{C}^n$ as the function space $\mathbb{C}^{\{1, 2, \dots, n\}}$, we can endow it with the inner product $\langle f, g \rangle = \sum_{k=1}^n f(k) \overline{g(k)}$, which gives the 2-norm $\|f\|_2 = \sqrt{\sum_{i=1}^n |f(i)|^2}$. The discrete Fourier basis is an orthogonal basis with respect to this inner product. Normalizing via division by $\sqrt{n}$ makes it an orthonormal basis. Thus, if we have $f \in \mathbb{C}^{\{1, 2, \dots, n\}}$ which we wish to write as $f = \sum_{j=1}^n a_j \chi_j$, then we can compute the coefficients a_j as

$$a_j = \frac{1}{n} \langle f, \chi_j \rangle = \frac{1}{n} \sum_{r=1}^n f(r) e^{-2\pi i j(r-1)/n}. \quad (1.2)$$

This is the discrete Fourier transform, as before. If we look at the norm, then

$$\|f\|_2^2 = \left\| \sum_{j=1}^n a_j \chi_j \right\|_2^2 = n \sum_{j=1}^n |a_j|^2 \implies \frac{1}{n} \|f\|_2^2 = \sum_{j=1}^n |a_j|^2. \quad (1.3)$$

This is known as *Plancherel's theorem*.

All of the above can be generalized to the continuous case; let $L^2[0, 1]$ be the space of square integrable functions on $[0, 1]$. If $(a_n)_{n \in \mathbb{Z}}$ is the Fourier series of f , then $\int_0^1 |f(x)|^2 dx = \sum_{n \in \mathbb{Z}} |a_n|^2$, which is the continuous analogue of Plancherel's theorem.

1.2 Metric and Normed Spaces

July 23rd.

The following theorem relates the notion of compactness, that every cover has a finite subcover, to many other properties of metric spaces.

Theorem 1.1. *Let (X, d) be a metric space. The following are equivalent:*

1. X is compact.
2. Every nested sequence of nonempty closed sets $F_1 \supseteq F_2 \supseteq \cdots$ has nonempty intersection, that is,

$$\bigcap_{n=1}^{\infty} F_n \neq \emptyset. \quad (1.4)$$

3. Every countable family of closed sets with the finite intersection property has nonempty intersection.
4. Every sequence in X has a convergent subsequence.
5. X is complete and totally bounded.

Proof. For $1 \implies 2$, let $\{F_n\}_{n=1}^{\infty}$ be a nested sequence of nonempty closed sets with $F_1 \supseteq F_2 \supseteq \cdots$. If $\bigcap_{n=1}^{\infty} F_n = \emptyset$, then $\{X \setminus F_n\}_{n=1}^{\infty}$ is an open cover of X . By compactness, there exists a finite subcover. Since the F_n are nested,

$$X = \bigcup_{i=1}^N (X \setminus F_i) = X \setminus F_N. \quad (1.5)$$

Hence $F_N = \emptyset$, a contradiction.

For $2 \implies 3$, let $\{F_n\}_{n=1}^{\infty}$ be a countable family of closed sets with the finite intersection property. Define

$$G_n = \bigcap_{i=1}^n F_i. \quad (1.6)$$

Then $G_1 \supseteq G_2 \supseteq \cdots$ is a nested sequence of nonempty closed sets. By 2, $\bigcap_{n=1}^{\infty} G_n \neq \emptyset$. Since

$$\bigcap_{n=1}^{\infty} G_n = \bigcap_{n=1}^{\infty} F_n, \quad (1.7)$$

the result follows.

For $3 \implies 4$, let S_0 be the set of elements in $(x_i)_{i=1}^{\infty}$, a sequence in X . Similarly, let S_1 be the set of elements in $(x_i)_{i=2}^{\infty}$, and define S_n similarly. Then

$$\overline{S_0} \supseteq \overline{S_1} \supseteq \cdots \supseteq \overline{S_n} \supseteq \cdots. \quad (1.8)$$

By 3, $\bigcap_{n=1}^{\infty} \overline{S_n} \neq \emptyset$, so there exists $x' \in \bigcap_{n=1}^{\infty} \overline{S_n}$. Now, $B(x', 1) \cap S_1$ cannot be empty, so pick x_{n_1} from this intersection. Again, $B(x', 1/2) \cap S_{n_1}$ cannot be empty, so pick x_{n_2} from this intersection. Continue inductively to obtain the x_{n_k} 's. One can verify that $x_{n_k} \rightarrow x'$.

For $4 \implies 5$, every Cauchy sequence has a convergent subsequence, and hence converges. Thus, X is complete. Suppose X is not totally bounded. Then there exists $\alpha > 0$ such that X has no finite covering by balls of radius α . Pick $x_1 \in X$. Since $B(x_1, \alpha) \neq X$, pick $x_2 \in X \setminus B(x_1, \alpha)$. Again,

$$B(x_1, \alpha) \cup B(x_2, \alpha) \neq X. \quad (1.9)$$

Similarly, pick x_3 outside the previous balls, and continue inductively. Since $d(x_m, x_n) \geq \alpha$ for $m \neq n$, there is no convergent subsequence, contradicting 4. Hence X is totally bounded.

For $5 \implies 1$, we prove the contrapositive. Assume X is not compact. If it is not totally bounded, we are done; further assume that it is totally bounded. Then there exists an open cover $\{U_\alpha\}_{\alpha \in \Lambda}$ with no finite subcover. Let $\varepsilon = 1/2$ and pick a finite ε -net. Pick x_1 such that $B(x_1, 1/2)$ is not covered by finitely many U_α . Since $B(x_1, 1/2)$ is totally bounded, pick $x_2 \in B(x_1, 1/2)$ such that $B(x_2, 1/4)$ cannot be covered by finitely many U_α . Inductively, pick $x_n \in B(x_{n-1}, 1/2^{n-1})$ such that $B(x_n, 1/2^n)$ cannot be covered by finitely many U_α . Then

$$d(x_n, x_m) \leq \sum_{k=m}^{n-1} \frac{1}{2^k}, \quad (1.10)$$

so (x_n) is a Cauchy sequence. Since X is complete, $x_n \rightarrow x$ for some $x \in X$. Let $x \in U_\alpha$. Then there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq U_\alpha$. Moreover, $d(x_n, x) < \varepsilon/2$ for n sufficiently large. Hence

$$B(x_n, 2^{-n}) \subseteq B(x_n, \varepsilon/2) \subseteq B(x, \varepsilon) \subseteq U_\alpha, \quad (1.11)$$

for all sufficiently large n , since $2^{-n} < \varepsilon/2$. This contradicts the choice of $B(x_n, 2^{-n})$. Hence X is not complete. ■

We now look at some examples of non-trivial function spaces; consider $C[0, 1]$, the space of complex-valued continuous functions on $[0, 1]$. This is a vector space over $\mathbb{R}$ with pointwise addition and scalar multiplication. We can define a norm on this space as $\|f\|_\infty = \sup_{x \in [0, 1]} |f(x)|$. This makes $C[0, 1]$ into a normed space.

Definition 1.2. Let X be a $\mathbb{K}$ -vector space. A *norm* on X is a function $\|\cdot\| : X \rightarrow [0, \infty)$ such that for all $x, y \in X$ and $\alpha \in \mathbb{K}$,

1. $\|x\| = 0$ if and only if $x = 0$ (positive definiteness).
2. $\|\alpha x\| = |\alpha| \|x\|$ (homogeneity).
3. $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality).

One may verify that $\|\cdot\|_\infty$ is indeed a norm on $C[0, 1]$.

Index

discrete Fourier basis, 1

discrete Fourier transform, 1

norm, 3

Plancherel's theorem, 2